

Discussion Module:

DSE Search, Intro, First Analyzer

Introduce the briefest amount of material possible to get our first DSE Search (text analytics/search) index in place, and run queries

- A DSE Search index in place to support synonyms. E.g., Dave, David, Davie, 'Dava Rooney'
- Useful for product/catalog searches; (mobile phone, cellular phone, smart phone), (car, auto, automobile, vehicle), (bike, bicycle, tricycle, moped), other

DSE Functional Areas: Core, Search

Discussion Lab:

Matching pairs – Match the attributes on the right with the areas on the left

Discussion Lab: Which DSE Functional Area



DSE Search

DSE Core



100-250+ ms Response



0-10 ms Response



Exact Match



Near Match (or exact match)



Hash Index



Tf-Idf (bitmap) Index

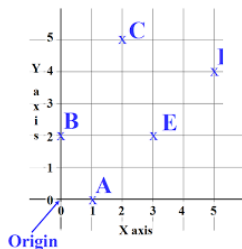


New Query Predicates

End of Discussion Lab:

DSE Search Index

- Sorting
- Facet (Group by)
- Wildcard (Text)
- Boosting
- Proximity
- (Other)



Integer, Double, String, ...

scalars

2

spatial

GeoSpatial

text

1

Sounds Like
Case Insensitive
Stemming
(Other)

Distance from point

3

Contained (other); Box,
rectangle, polygon

(Time Series)



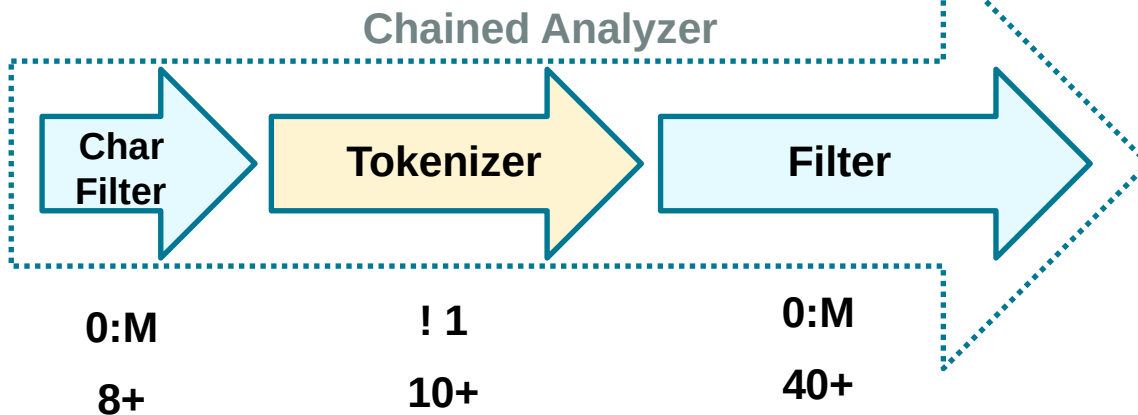
DSE Search: the Analyzer

Analysis Phase: Index|
Query|(Both)




Document
(Column Value)


Document
(Column Value)




	Document 1	Document 2	Document 3	Document 4	Document 5	Document 6	Document 7	Document 8
Term(s) 1	10	0	1	0	0	0	0	2
Term(s) 2	0	2	0	0	0	18	0	2
Term(s) 3	0	0	0	0	0	0	0	2
Term(s) 4	6	0	0	4	6	0	0	0
Term(s) 5	0	0	0	0	0	0	0	2
Term(s) 6	0	0	1	0	0	1	0	0
Term(s) 7	0	1	8	0	0	0	0	0
Term(s) 8	0	0	0	0	0	3	0	0

Word Vector (Passage Vector)

Document Vector



	Document 1	Document 2	Document 3	Document 4	Document 5	Document 6	Document 7	Document 8
Term(s) 1	10	0	1	0	0	0	0	2
Term(s) 2	0	2	0	0	0	18	0	2
Term(s) 3	0	0	0	0	0	0	0	2
Term(s) 4	6	0	0	4	6	0	0	0
Term(s) 5	0	0	0	0	0	0	0	2
Term(s) 6	0	0	1	0	0	1	0	0
Term(s) 7	0	1	8	0	0	0	0	0
Term(s) 8	0	0	0	0	0	3	0	0

Word Vector (Passage Vector)

Document Vector

Catalog of Available Analyzer Options:



- Chained Analyzers-
 - Char Filters (8+), https://lucene.apache.org/solr/guide/6_6/charfilterfactories.html
 - Tokenizers (10+), https://lucene.apache.org/solr/guide/6_6/tokenizers.html
 - Filters (40+), https://lucene.apache.org/solr/guide/6_6/filter-descriptions.html
- Unchained Analyzers (6+30),
<https://www.solr-start.com/info/analyzers/>

DSE Search,: Field Types, Fields

2
field

(10+)

name

type (refers to field
type class name)

docValues

multiValued

indexed

(Other)

field
1
types

(8+, not including the data types)

name

class (data type)

docValues ?

positionIncrementGap

(Other)



Catalog of Available Field Type / Field Options:



Field Type attributes/properties (8+ not including data types) :

[https://
lucene.apache.org/solr/guide/6_6/field-type-definitions-and-properties.html](https://lucene.apache.org/solr/guide/6_6/field-type-definitions-and-properties.html)

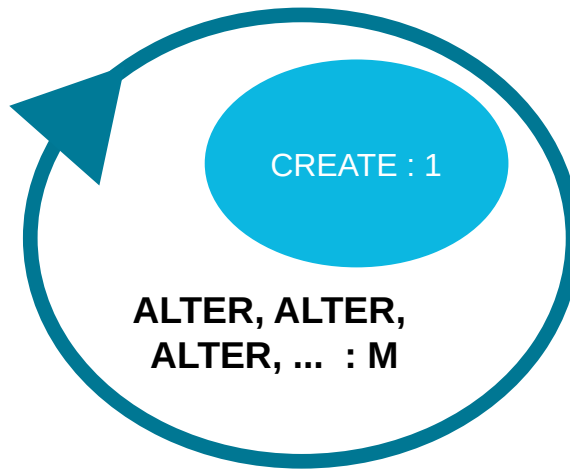
Field attributes/properties (20+) :

[https://
lucene.apache.org/solr/guide/6_6/field-properties-by-use-case.html](https://lucene.apache.org/solr/guide/6_6/field-properties-by-use-case.html)

https://lucene.apache.org/solr/guide/6_6/defining-fields.html

Field Types <- Analyzers (Indexes), Fields: How are These Created-

- dsetool: schema.xml, solrconfig.xml
- CQL
 - (CREATE TABLE)
 - CREATE SEARCH INDEX ...
 - **ALTER SEARCH INDEX ...**
 - RELOAD SEARCH INDEX ...
 - REBUILD SEARCH INDEX ...
 - DESCRIBE ACTIVE | PENDING SEARCH INDEX SCHEMA | CONFIG ..
 - DROP SEARCH INDEX ...



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- Gremlin (for Graph only)

End to End Example:

Synonym, case sensitive

- DSE cluster, with Search enabled
 - Authority to make tables, other
 - CQLSH/Studio access-
-
- Will leave it case sensitive, but will offer clues for actual Practice Lab that follows



Synonym: CREATE KEYSPACE

```
DROP KEYSPACE ks_7443;  
CREATE KEYSPACE ks_7443 WITH REPLICATION =  
    {'class': 'SimpleStrategy', 'replication_factor': 1};  
USE ks_7443;
```



Synonym: CREATE TABLE

```
CREATE TABLE t1
(
  col1      TEXT,
  col2      TEXT,
  col3      TEXT,
  col4      TEXT,
  col5      TEXT,
  col6      TEXT,
  col7      TEXT,
  col8      TEXT,
  PRIMARY KEY ((col1, col2)) );
```



Synonym: (Add Data)

```
INSERT INTO t1
```

```
  (col1, col2, col3, col4,  
   col5, col6, col7, col8)
```

```
VALUES ('aaa', 'aaa', 'Davie' ,  
       'aaa', 'aaa', 'aaa', 'aaa', 'aaa');
```

```
INSERT INTO t1
```

```
  (col1, col2, col3, col4,  
   col5, col6, col7, col8)
```

```
VALUES ('bbb', 'bbb', 'Elizabeth',  
       'bbb', 'bbb', 'bbb', 'bbb', 'bbb');
```



Synonym: CREATE SEARCH INDEX

```
CREATE SEARCH INDEX ON t1 WITH COLUMNS col1,  
  * { excluded : true };
```

>>Warnings :

>>Please remember this operation is DC specific and should be

>>repeated on each desired DC.



Synonym: DESCRIBE TABLE

DESCRIBE TABLE t1;

```
>>CREATE TABLE ks_7443.t1 (  
>>  col1 text,  
>>  col2 text,  
>>  ...  
>>  col8 text,  
>>  solr_query text,  
>>  PRIMARY KEY ((col1, col2))  
>>) WITH bloom_filter_fp_chance = 0.01  
>>  AND caching = ...  
>>  AND comment = "  
>>  AND compaction = ...  
>>  ...  
>>  AND speculative_retry = '99PERCENTILE';  
>>CREATE CUSTOM INDEX ks_7443_t1_solr_query_index  
>> ON ks_7443.t1 (solr_query) USING  
>> 'com.datastax.bdp.search.solr.Cql3SolrSecondaryIndex';
```


Synonym: DESCRIBE ... SEARCH INDEX

DESCRIBE PENDING SEARCH INDEX SCHEMA ON t1;

```
>> <?xml version="1.0" encoding="UTF-8" standalone="no"?>
>> <schema name="autoSolrSchema" version="1.5">
>>   <types>
>>     <fieldType class="org.apache.solr.schema.StrField"
>>       name="StrField"/>
>>   </types>
>>   <fields>
>>     <field indexed="true" multiValued="false"
>>       name="col1" type="StrField"/>
>>     <field indexed="true" multiValued="false"
>>       name="col2" type="StrField"/>
>>   </fields>
>>   <uniqueKey>(col1,col2)</uniqueKey>
>> </schema>
```



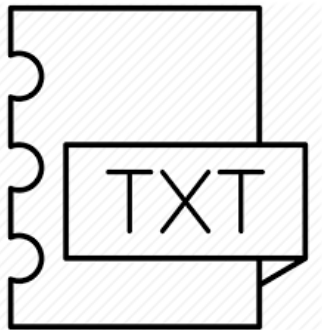
col2 ??

StrField -vs- TextField

(>= DSE 6.0)

Synonym: Make an ASCII Text Field, Submit

```
dsetool write_resource ks_7443.t1 \  
  name=7443_my_synonyms.txt \  
  file=7443_my_synonyms.txt
```



```
#  
# My synonyms file !!  
#  
Robert, Bob, Bobbie  
Dave, David, Davie, Dava Rooney  
Charles, Chuck, Charlie, Barkley  
Elizabeth, Beth, Liz, Lizzie, Tina
```



**Dropped if
you drop
the index**

Synonym: Progressive Examples, surplus steps-

ALTER SEARCH INDEX SCHEMA ON t1

```
ADD fieldType[ @name='TextField_5',  
@class='solr.TextField'];
```



ALTER SEARCH INDEX SCHEMA ON t1

```
ADD fieldType[@name='TextField_6',  
@class='solr.TextField',  
@termVectors='false',  
@termPositions='false',  
@termOffsets='false',  
@omitNorms='true'];
```



Not using Highlighting or other
functionality that depends on it

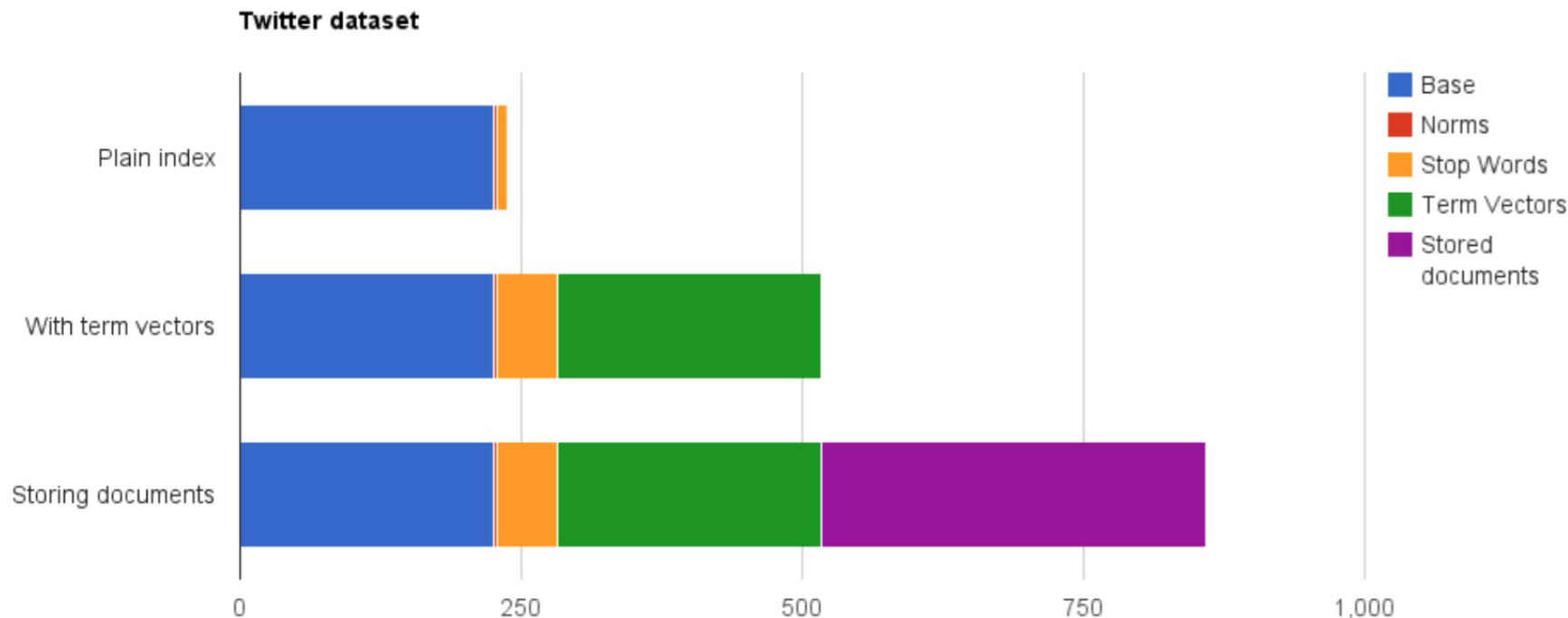


Not using term Boosting



**Saves
disk space,
other**

Apache Solr, Impact on Disk Space, termVectors, more ..



Source: Slide 36, <https://www.slideshare.net/DataStax/solr47-f-inal?https://data.world/datasets/twitter>

Synonym: Field Type (minimum)

```
ALTER SEARCH INDEX SCHEMA ON t1
```

```
  ADD types.fieldType[@name='TextField_7',  
    @class='solr.TextField'] with    '{"analyzer":{"tokenizer":  
{"class":"solr.StandardTokenizerFactory"},  
  "filter":{"class":"solr.SynonymFilterFactory","synonyms":"7443_my_synonyms.txt"}  
}}';
```

// File not loaded ?

// InvalidRequest: Error from server: code=2200 [Invalid query] message=

// "The search index schema is not valid because: java.io.IOException: Could not find: 47_my_synonyms.txt"

Synonym: Field Type (actual)

ALTER SEARCH INDEX SCHEMA ON t1

```
ADD types.fieldType[@name='TextField_7',  
@class='solr.TextField',@termVectors='false',@termPositions='false',  
@termOffsets='false',@omitNorms='true'] with  
'{"analyzer":{"tokenizer":{"class":"solr.StandardTokenizerFactory"},  
"filter":{"class":"solr.SynonymFilterFactory","synonyms":"7443_my_synonyms.txt"}}}';
```

Synonym: Add Field, and Deploy

```
ALTER SEARCH INDEX SCHEMA ON t1
```

```
  ADD fields.field[@name='col3', @type='TextField_7',  
    @indexed='true', @multiValued='false', @stored='true'];
```

```
RELOAD  SEARCH INDEX ON t1;
```

// Warnings :

// Please remember this operation is DC specific and should be repeated on each desired DC.

```
REBUILD SEARCH INDEX ON t1;
```

// Warnings :

// Please remember this operation is DC specific and should be repeated on each desired DC.

Synonym: Test, CQL SELECT

```
SELECT col1, col2, col3 FROM t1 WHERE solr_query = '{ "q" : "col3:Dave" }' ;
```

```
>>   col1 | col2 | col3
```

```
>>   aaa |  aaa | Davie
```

```
>> (1 rows)
```

```
SELECT col1, col2, col3 FROM t1 WHERE solr_query = '{ "q" : "col3:(Dava Rooney)" }' ;
```

```
>>   col1 | col2 | col3
```

```
>>   aaa |  aaa | Davie
```

```
>> (1 rows)
```

```
SELECT col1, col2, col3 FROM t1 WHERE solr_query = '{ "q" : "col3:dave" }' ;
```

```
>>   col1 | col2 | col3
```

```
>> (0 rows)
```


Synonym: Debugging (page 1 of 2)

DESCRIBE ACTIVE SEARCH INDEX SCHEMA ON t1;

```
>> <?xml version="1.0" encoding="UTF-8" standalone="no"?>
```

```
>> <schema name="autoSolrSchema" version="1.5">
```

```
>>   <types>
```

```
>>     <fieldType class="org.apache.solr.schema.StrField" name="StrField"/>
```

```
>>     <fieldType class="solr.TextField" name="TextField_7" omitNorms="true"
```

```
>>       termOffsets="false" termPositions="false" termVectors="false">
```

```
>>       <analyzer>
```

```
>>         <filter class="solr.SynonymFilterFactory"
```

```
>>           synonyms="7443_my_synonyms.txt"/>
```

```
>>       <tokenizer class="solr.StandardTokenizerFactory"/>
```

```
...
```

Synonym: Debugging (page 2 of 2)

...

```
>> </analyzer>
>> </fieldType>
>> </types>
>> <fields>
>>   <field indexed="true" multiValued="false" name="col1" type="StrField"/>
>>   <field indexed="true" multiValued="false" name="col2" type="StrField"/>
>>   <field indexed="true" multiValued="false" name="col3" stored="true"
>>     type="TextField_7"/>
>> </fields>
>> <uniqueKey>(col1,col2)</uniqueKey>
>> </schema>
```

End to End Example:

Synonym, case sensitive,
Review

- CREATE SEARCH INDEX ... (PK)
- Synonym -> dsetool write_resource
- ALTER ... ADD types.fieldType ...
- ALTER ... ADD fields.field ...
- (deploy)
- Case insensitive ? (Next page)



Two code fragments: Lower case, and multiple filters

```
ALTER SEARCH INDEX SCHEMA ON t2
```

```
  ADD fieldType[@name='TextField44',class='solr.TextField']  
  WITH '{"analyzer":{"tokenizer":{"class":"solr.StandardTokenizerFactory"},  
    "filter":{"class":"solr.LowerCaseFilterFactory"}}}';
```

```
ALTER SEARCH INDEX SCHEMA ON t2
```

```
  ADD fieldType[@name='TextField41',class='solr.TextField']  
  WITH '{"analyzer":{"tokenizer":{"class":"solr.StandardTokenizerFactory"},  
    "filter":[{"class":"solr.LowerCaseFilterFactory"},  
      {"class":"solr.EnglishMinimalStemFilterFactory"}]}}';
```



End of Module:



Solutions:

Discussion Lab: Which DSE Functional Area

